

Thm. Let $\{a_n\}$ be a series, and take $R = (\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}})^{-1}$.

Then, the series $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely if $|x| < R$,
diverges if $|x| > R$.

Therefore, we can define $f: (-R, R) \rightarrow \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n x^n$.

For any $0 < \epsilon < R$, f converges uniformly on $[-R+\epsilon, R-\epsilon]$ and
 f is differentiable on $(-R, R)$, with

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

proof. The first assertion is given by the root test.

And, the second assertion is given by the Weierstrass-M test, specifically,
for all $n \in \mathbb{N}$ and $x \in [-R+\epsilon, R-\epsilon]$, $|a_n x^n| = |a_n| |x|^n \leq |a_n| \cdot (R-\epsilon)^n$, and

$$\sum_{n=0}^{\infty} |a_n| \cdot (R-\epsilon)^n$$

Revised in the next page.

converges, by the first. To show the third assertion, observe that

$$\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \limsup_{n \rightarrow \infty} |n a_n|^{\frac{1}{n}}$$

Because: since $x_n = n^{\frac{1}{n}} - 1$ satisfies $(1+x_n)^n = n \geq \frac{n(n-1)}{2} \cdot x_n^2$, thus

$$x_n^2 \leq \frac{2}{n-1} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Thus $n^{\frac{1}{n}} \rightarrow 1$ as $n \rightarrow \infty$. Given $\epsilon > 0$, there is $N_1 \in \mathbb{N}$ s.t

$$n \geq N_1 \Rightarrow |a_n|^{\frac{1}{n}} < \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} + \frac{\epsilon}{2}$$

Meanwhile, since $n^{\frac{1}{n}} \rightarrow 1$ as $n \rightarrow \infty$, take $N_2 \in \mathbb{N}$ s.t

$$n \geq N_2 \Rightarrow n^{\frac{1}{n}} < 1 + r, \text{ where } r = \frac{A+\epsilon}{A+\frac{\epsilon}{2}} - 1, \quad A = \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}.$$

Then,
 $n \geq \max\{N_1, N_2\} \Rightarrow n^{\frac{1}{n}} \cdot |a_n|^{\frac{1}{n}} \leq \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} + \epsilon.$

Moreover, there are infinitely many $n \in \mathbb{N}$ s.t $\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} - \frac{\epsilon}{2} < |a_n|^{\frac{1}{n}}$,

we can choose similarly, 'infinitely many $n \in \mathbb{N}$ s.t $1 - t < n^{\frac{1}{n}}$, $t = 1 - \frac{A-\epsilon}{A-\frac{\epsilon}{2}}$

Now proved. Using this, $\sum n a_n x^{n-1}$ converges uniformly in $(-R, R)$,

we obtain the result.

Thm. Let $\{a_n\}$ be a sequence in $\mathbb{R}$ with $R = \left(\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}\right)^{-1}$.

Then, the power series

$$\sum_{n=0}^{\infty} a_n x^n \quad \begin{array}{l} \text{Converges for all } |x| < R, \\ \text{diverges for all } |x| > R. \end{array} \quad \text{By root test.}$$

Therefore, we can define

$$f: (-R, R) \rightarrow \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n x^n.$$

The function f satisfies:

1) For any $0 < \epsilon < R$, f is uniformly continuous on $[-R+\epsilon, R-\epsilon]$

2) $f \in C^\infty(-R, R)$ with $f'(x) = \sum_{n=1}^{\infty} n \cdot a_n x^{n-1}$.

Proof. Let $x \in (-R, R)$ be given. Then, for each $n \in \mathbb{N}$,

$$\sqrt[n]{|a_n \cdot x^n|} = \sqrt[n]{|a_n| \cdot |x|^n} = |a_n|^{\frac{1}{n}} \cdot |x|$$

Therefore, by the root test, $\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} \cdot |x| = |x| \cdot \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} < R \cdot R^{-1} = 1$,
the given series converges. ($\sqrt[n]{|b_n|} < 1 \Rightarrow a_n < \beta^n$ for some $\beta < 1$.)

To show 1), let $0 < \epsilon < R$ be given. For each $n \in \mathbb{N}$, and $x \in [-R+\epsilon, R-\epsilon]$,

$$|a_n x^n| = |a_n| \cdot |x|^n \leq |a_n| \cdot |R-\epsilon|^n \quad \text{and} \quad \sum_{n=0}^{\infty} |a_n| \cdot |R-\epsilon|^n < \infty,$$

therefore by the Weierstrass M-test, $\sum a_n x^n$ converges uniformly to f ,
so the f is continuous on the compact set $[-R+\epsilon, R-\epsilon]$, so uniformly conti.

To show 2), put $f_N(x) = \sum_{n=0}^N a_n x^n$. Then, $f'_N(x) = \sum_{n=1}^N n a_n x^{n-1}$.

Since $\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \limsup_{n \rightarrow \infty} |n a_n|^{\frac{1}{n}}$, $\lim_{N \rightarrow \infty} f'_N(x)$ converges in $(-R, R)$.

Moreover, f_N converges to f in $(-R, R)$ and

for any $0 < \epsilon < R$, f'_N converges uniformly, because given $x \in [-R+\epsilon, R-\epsilon]$,

$$|n \cdot a_n \cdot x^{n-1}| \leq n \cdot a_n \cdot |R-\epsilon|^{n-1} \quad \text{and} \quad \sum_{n=1}^{\infty} n a_n |R-\epsilon|^{n-1} \text{ converges by the root test,}$$

therefore the Weierstrass M-test gives f'_N converges uniformly.

Now, given $x \in (-R, R)$, we can take $\epsilon > 0$ such that $x < R-\epsilon < R$, thus

$$\lim_{N \rightarrow \infty} f'_N(x) = \sum_{n=1}^{\infty} n \cdot a_n \cdot x^{n-1} = f'(x).$$

by the theorem for differentiation

Thm. Suppose that $f: (-R, R) \rightarrow \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n \cdot x^n$ is well-defined.

Given $c \in (-R, R)$, the representation

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} \cdot (x-c)^n \quad (|x-c| < R-|c|)$$

is valid.

Proof. Given $c \in (-R, R)$

$$f(x) = \sum_{n=0}^{\infty} a_n \cdot x^n = \sum_{n=0}^{\infty} a_n \cdot [x-c+c]^n = \sum_{n=0}^{\infty} a_n \cdot \sum_{m=0}^n \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m$$

Meanwhile, since $|x-c| + |c| < R$,

$$\begin{aligned} \sum_{n=0}^{\infty} \sum_{m=0}^n |a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m| &= \sum_{n=0}^{\infty} \sum_{m=0}^n |a_n| \cdot \binom{n}{m} \cdot |c|^{n-m} \cdot |x-c|^m \\ &= \sum_{n=0}^{\infty} |a_n| \cdot [|x-c| + |c|]^n < \infty. \quad \dots (*) \end{aligned}$$

Put
$$A_{nm} = \begin{cases} a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m & \text{if } n \geq m \\ 0 & \text{if } n < m. \end{cases}$$
 Then,

$\sum_{m=0}^{\infty} |A_{nm}|$ converges for all $n \in \mathbb{N}$, (because for fixed n , $\sum_{m=0}^{\infty} |A_{nm}|$ is just finite sum,

actually, $\sum_{m=0}^{\infty} |A_{nm}| = \sum_{m=0}^n |A_{nm}|$.) and moreover, $\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} |A_{nm}|$ converges by (*). Now

$$\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} A_{nm} = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} A_{nm} = \sum_{m=0}^{\infty} \sum_{n=m}^{\infty} A_{nm} = \sum_{m=0}^{\infty} \left[\sum_{n=m}^{\infty} \binom{n}{m} \cdot a_n \cdot c^{n-m} \right] \cdot (x-c)^m$$

Finally, $\frac{1}{m!} \cdot f^{(m)}(x) = \sum_{n=m}^{\infty} \binom{n}{m} \cdot a_n \cdot x^{n-m}$ gives the result, just substituting $x=c$. $\square$

★ This Proof is revised at 2026.8.2.

Thm. Suppose that $f: (-R, R) \rightarrow \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n \cdot x^n$ is well-defined.

Given $c \in (-R, R)$, the equality holds:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} \cdot (x-c)^n \quad (x \in \mathbb{B}_{R-|c|}(c)).$$

Proof. Observe that

$$f(x) = \sum_{n=0}^{\infty} a_n \cdot [(x-c) + c]^n = \sum_{n=0}^{\infty} a_n \cdot \sum_{m=0}^n \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m.$$

And, since $x \in \mathbb{B}_{R-|c|}(c) \Leftrightarrow |x-c| + |c| < R$,

$$\sum_{n=0}^{\infty} \sum_{m=0}^n |a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m| = \sum_{n=0}^{\infty} |a_n| \cdot [|x-c| + |c|]^n < \infty.$$

So, $\sum_{m=0}^{\infty} |A_{nm}|$ converges for each n , where $A_{nm} = a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m$.

We can change the limits

$$f(x) = \sum_{m=0}^{\infty} \left(\sum_{n=m}^{\infty} \binom{n}{m} a_n \cdot c^{n-m} \right) (x-c)^m = \sum_{m=0}^{\infty} \frac{f^{(m)}(c)}{m!} \cdot (x-c)^m.$$

Thm. Given series $\sum a_n$ and $\sum b_n$, define $C_n = \sum_{k=0}^n a_k \cdot b_{n-k}$.

If $\sum_{n=0}^{\infty} a_n$ Converges absolutely and $\sum_{n=0}^{\infty} b_n$ Converges, then

$$\sum_{n=0}^{\infty} C_n = \left(\sum_{n=0}^{\infty} a_n \right) \cdot \left(\sum_{n=0}^{\infty} b_n \right).$$

Proof. Denote

$$A_n = \sum_{k=0}^n a_k, \quad B_n = \sum_{k=0}^n b_k, \quad B = \sum_{n=0}^{\infty} b_n \quad \text{and} \quad \beta_n = B_n - B$$

Then,

$$\begin{aligned} \sum_{k=0}^n a_k &= a_0 b_0 + (a_0 b_1 + a_1 b_0) + \dots + (a_0 b_n + a_1 b_{n-1} + \dots + a_n b_0) \\ &= a_0 B_n + a_1 B_{n-1} + \dots + a_n B_0 \\ &= a_0 (B + \beta_n) + a_1 (B + \beta_{n-1}) + \dots + a_n (B + \beta_0) \\ &= A_n B + a_0 \beta_n + a_1 \beta_{n-1} + \dots + a_n \beta_0. \end{aligned}$$

Now, enough to show that

$$\gamma_n = a_0 \beta_n + a_1 \beta_{n-1} + \dots + a_n \beta_0$$

Converges to zero.

Let $\epsilon > 0$ be given. Since $\beta_n \rightarrow 0$ as $n \rightarrow \infty$, there exists $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow |\beta_n| \leq \epsilon$.

Now, for $n \geq N$,

$$\begin{aligned} |\gamma_n| &\leq |\beta_0 a_n + \dots + \beta_N a_{n-N}| + |\beta_{N+1} a_{n-N+1} + \dots + \beta_n a_0| \\ &\leq |\beta_0 a_n + \dots + \beta_N a_{n-N}| + \epsilon \cdot (|a_{N+1}| + \dots + |a_0|) \\ &\leq |\beta_0 a_n + \dots + \beta_N a_{n-N}| + \epsilon \cdot \sum_{k=0}^{\infty} |a_k| \quad \leftarrow \end{aligned}$$

Using this, Converges absolutely.

Take $\limsup$, then

$$\begin{aligned} \limsup_{n \rightarrow \infty} |\gamma_n| &\leq \limsup_{n \rightarrow \infty} |\beta_0 a_n + \dots + \beta_N a_{n-N}| + \epsilon \sum_{k=0}^{\infty} |a_k| \\ &= \epsilon \cdot \sum_{k=0}^{\infty} |a_k|. \end{aligned}$$

Consequently, $\gamma_n \rightarrow 0$ as $n \rightarrow \infty$, so that we obtain the result.

Abel.

Thm. Let $f: [-1, 1) \rightarrow \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n x^n$ be defined.

If $\sum_{n=0}^{\infty} a_n$ converges, then $\lim_{x \rightarrow 1^-} f(x) = \sum_{n=0}^{\infty} a_n$.

Proof. Put $A_n = \sum_{k=0}^n a_k$, and $A = \lim_{n \rightarrow \infty} A_n = \sum_{n=0}^{\infty} a_n$. As a Cauchy product,

$$\frac{1}{1-x} f(x) = \sum_{n=0}^{\infty} A_n \cdot x^n, \text{ so } f(x) = (1-x) \cdot \sum_{n=0}^{\infty} A_n x^n \quad (-1 < x < 1)$$

and since $(1-x) \cdot \sum_{n=0}^{\infty} x^n = 1$, so

$$f(x) - A = (1-x) \sum_{n=0}^{\infty} A_n x^n - A \cdot (1-x) \sum_{n=0}^{\infty} x^n$$

$$= (1-x) \left(\sum_{n=0}^{\infty} A_n x^n - A \cdot \sum_{n=0}^{\infty} x^n \right)$$

$$= (1-x) \cdot \left(\sum_{n=0}^{\infty} (A_n - A) \cdot x^n \right), \quad (-1 < x < 1).$$

Given $\varepsilon > 0$, take $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow |A_n - A| < \varepsilon$, and put $M = \max \{|A_n - A| \mid n = 0, \dots, N\}$.

$$\text{Now, } |f(x) - A| \leq \left| (1-x) \cdot \sum_{n=0}^N (A_n - A) x^n \right| + \left| (1-x) \cdot \sum_{n=N}^{\infty} (A_n - A) x^n \right|$$

$$\leq (1-x) \cdot MN + \varepsilon$$

Therefore, $\lim_{x \rightarrow 1^-} f(x) = A$.

The Uniqueness Theorem of the power series.

Thm. Suppose that the function of power series

$$f: (-R, R) \rightarrow \mathbb{R}: x \mapsto \sum_{n=0}^{\infty} a_n \cdot x^n$$

is well-defined.

If $Z(f)$ has a limit point in $(-R, R)$, then $a_n = 0$ for all $n \geq 0$.

Proof. Let $A = \{x \in (-R, R) \mid x \text{ is the limit point of } Z(f)\}$. (A is non-empty, by the hypo-)

Given $x_0 \in A$, the representation

$$f(x) = \sum_{n=0}^{\infty} a_n \cdot (x - x_0)^n \text{ is valid in } |x - x_0| < R - |x_0|.$$

To show that $a_n = 0$ for all $n \geq 0$, let k be the smallest non-negative integer s.t. $a_k \neq 0$.

Then, we can write $f(x) = (x - x_0)^k \cdot g(x)$ where $g(x) = \sum_{m=0}^{\infty} a_{k+m} \cdot (x - x_0)^m$. ($|x - x_0| < R - |x_0|$).

And, the g is continuous at x_0 , because the each term is continuous and converges uniformly.

And, $g(x_0) \neq 0$ by the construction. So, for some $\delta > 0$, $g(x) \neq 0$ in $|x - x_0| < \delta$.

So, $f(x) \neq 0$ in $0 < |x - x_0| < \delta$, it contradicts to x_0 is the limit point of $Z(f)$.

Therefore, $f(x) = 0$ for all $|x - x_0| < R - |x_0|$, it means that

$$x_0 \in \bigcup_{R-|x_0|} B_{R-|x_0|}(x_0) \subseteq Z(f), \text{ and so } B_{R-|x_0|} \cap (-R, R) \subseteq A, \text{ so } A \text{ is open.}$$

And, the set of limit points is closed, A is the non-empty closed set of $(-R, R)$.

But, since $(-R, R)$ is connected, the only possibility is $A = (-R, R)$.

Moreover, since f is continuous, $A \subseteq Z(f)$, so $Z(f) = (-R, R)$. $\square$